Relational Time and Covariant Foliations in Relativistic Quantum Mechanics

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ABSTRACT

General covariance in General Relativity eliminates any preferred external time parameter, causing the canonical Hamiltonian of closed gravitational systems to vanish identically under the Wheeler–DeWitt constraint. Conversely, standard Quantum Mechanics relies on a rigid external parameter to generate unitary state evolution. This work reconciles both frameworks by coupling Arnowitt–Deser–Misner (ADM) 3+1 spacetime foliations and Zero-Angular-Momentum Observer (ZAMO) congruences with the Page–Wootters conditional probability formalism and Tomonaga–Schwinger multi-fingered time evolution. We synthesize these structures into dimensionless relational metrics—the ZAMO Dilation Factor (ρ_τ), the Clock Rate Delta ($\Delta\rho_\tau$), the ZAMO Back-Reaction Factor (H_Q), Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$), and the stretched-horizon membrane regulator (Ξ_{QG})—demonstrating how unitary quantum phase evolution remains well-defined across curved, rotating ergospheres.

Keywords: Problem of Time; ADM 3+1 Formalism; Wheeler–DeWitt Equation; Page–Wootters Mechanism; Tomonaga–Schwinger Equation; Zero-Angular-Momentum Observer (ZAMO); Relational Quantum Dynamics.

1. Canonical General Relativity and the Wheeler–DeWitt Constraint

In non-relativistic Quantum Mechanics, quantum states evolve parametrically against an absolute Newtonian clock t via the Schrödinger equation ($i\hbar\,\partial_t\,|\psi\rangle = \hat{H}|\psi\rangle$). In General Relativity, diffeomorphism invariance treats spacetime coordinates as arbitrary labels, eliminating any preferred global time.

1.1 Arnowitt–Deser–Misner (ADM) 3+1 Spacetime Foliation

Assuming a globally hyperbolic spacetime $\mathcal{M} \cong \mathbb{R} \times \Sigma$, we foliate the four-dimensional manifold into non-intersecting spacelike hypersurfaces Σ_t parameterized by coordinate time t (Arnowitt, Deser, & Misner, 1962):

$$ds^2 = -N^2\,c^2\,dt^2 + h_{ij}\,(dx^i + N^i\,c\,dt)(dx^j + N^j\,c\,dt)$$

- **N(t, x):** The *lapse function*, governing proper time advancement normal to Σ_t ($d\tau = N\,dt$).

- **$N^i(t, x)$:** The *shift vector*, measuring how spatial coordinates drift relative to the normal congruence (reducing to azimuthal frame-dragging $N^\varphi = -\omega_{FD}$ in stationary axisymmetric spacetimes).
- **h_{ij}:** The induced Riemannian three-metric on each spatial slice Σ_t.

1.2 Dirac–Bergmann First-Class Constraints and the Frozen Formalism

Performing a Legendre transformation of the Einstein–Hilbert action shows that N and N^i act as Lagrange multipliers enforcing four first-class constraints (DeWitt, 1967):

$$\mathcal{H}_\perp = (16\pi G / c^4)\, G_{ijkl}\, \pi^{ij} \pi^{kl} - (c^4 \sqrt{h} / 16\pi G)({}^{(3)}R - 2\Lambda) + \mathcal{H}_{\text{matter}} \approx 0, \qquad \mathcal{H}_i = -2\, D_j\, \pi^j{}_i + \mathcal{H}_{i,\text{matter}} \approx 0$$

where $G_{ijkl} = (1 / 2\sqrt{h})(h_{ik}h_{jl} + h_{il}h_{jk} - h_{ij}h_{kl})$ is the DeWitt supermetric and π^{ij} are canonical momenta conjugate to h_{ij}. Upon canonical quantization ($\hat{\mathcal{H}}_\perp |\Psi\rangle = 0$), the total Hamiltonian annihilates the physical state, yielding the timeless **Wheeler–DeWitt equation** in which external coordinate evolution vanishes.

2. Relational Dynamics: Page–Wootters and Tomonaga–Schwinger Evolution

To recover unitary quantum evolution without an external clock, time must be defined relationally—either by conditioning one quantum subsystem against a physical clock observable, or by evolving quantum fields locally across arbitrary curved hypersurfaces.

2.1 The Page–Wootters Conditional Probability Formalism

Partition the total Hilbert space into a physical clock subsystem $\mathcal{H}_C$ and a target quantum system $\mathcal{H}_S$ ($\mathcal{H}_{tot} = \mathcal{H}_C \otimes \mathcal{H}_S$), governed by the global constraint $(\hat{H}_C \otimes \hat{1}_S + \hat{1}_C \otimes \hat{H}_S)|\Psi\rangle = 0$ (Page & Wootters, 1983). Defining clock states $|\tau\rangle = \exp(-i\, \hat{H}_C\, \tau / \hbar)|0\rangle$, the **conditional density matrix** of the system given clock reading τ is:

$$\rho_S(\tau) = \mathrm{Tr}_C[\, (|\tau\rangle\langle\tau| \otimes \hat{1}_S)\, |\Psi\rangle\langle\Psi|\,] \,/\, \mathrm{Tr}[\, (|\tau\rangle\langle\tau| \otimes \hat{1}_S)\, |\Psi\rangle\langle\Psi|\,]$$

Differentiating $\rho_S(\tau)$ with respect to the internal clock reading τ recovers exact unitary von Neumann evolution:

$$i\hbar\, (d\rho_S(\tau) / d\tau) = [\hat{H}_S, \rho_S(\tau)]$$

2.2 Tomonaga–Schwinger Multi-Fingered Time Functional Equation

For relativistic quantum fields on curved backgrounds, a single global time t violates local Lorentz covariance. Replacing ∂_t with local functional differentiation $\delta / \delta\sigma(x)$ across arbitrary spacelike hypersurfaces $\sigma(x)$ yields the **Tomonaga–Schwinger equation** (Tomonaga, 1946; Schwinger, 1948):

$$i\hbar \, (\delta|\Psi[\sigma]\rangle / \delta\sigma(x)) = \mathscr{H}_{int}(x) \, |\Psi[\sigma]\rangle$$

Microcausality ($[\mathscr{H}_{int}(x), \mathscr{H}_{int}(y)] = 0$ for spacelike-separated points) guarantees integrability, ensuring that physical transition amplitudes depend only on the initial and final hypersurfaces rather than the intermediate foliation path.

3. Dimensionless Relational Metrics and Ergosphere Regulation

To operationalize relational time across curved, rotating spacetimes, we anchor local quantum phase evolution to Zero-Angular-Momentum Observer (ZAMO) worldlines (Bardeen, Press, & Teukolsky, 1972).

3.1 ZAMO Dilation Factor (ρ_τ) and Clock Rate Delta ($\Delta\rho_\tau$)

In stationary axisymmetric Kerr–Newman geometries, static observers become spacelike inside the ergosphere ($-g_{tt} \leq 0$). Selecting the future-directed ZAMO unit normal $n^\mu = (1/N, 0, 0, \omega_{FD}/N)$ with lapse $N^2(\theta) = \Sigma\Delta / A_{BL}$ keeps timelike foliations well-defined down to the outer event horizon (r_+). Comparing proper time along a target worldline ($d\tau_{pre} = \sqrt{N^2}_{pre}(\theta) / \gamma_{pre} \, dt$) against a physical observer clock ($d\tau_{obs} = \sqrt{(1 - \mathcal{C}_{obs})} / \gamma_{obs} \, dt$) defines the relational clock exchange rate:

$$\rho_\tau \equiv d\tau_{obs} / d\tau_{pre} = [\, \gamma_{pre} / \sqrt{N^2}_{pre}(\theta) \,] / [\, \gamma_{obs} / \sqrt{(1 - \mathcal{C}_{obs})} \,], \qquad \Delta\rho_\tau = \rho_\tau - 1$$

This eliminates unphysical coordinate time *dt* completely, replacing it with the exact ratio of proper ticks between two physical clocks.

3.2 ZAMO Back-Reaction Factor (H_Q)

Relating the redshifted asymptotic oscillation frequency $f_\infty = f_{local} \sqrt{N^2}_{pre}(\theta)$ of a quantum wavepacket to its rest-frame Compton frequency $f_0 = m_0c^2 / h$ defines the dimensionless **ZAMO Back-Reaction Factor**:

$$H_Q \equiv \sqrt{N^2}_{pre}(\theta) / \gamma_{pre} \in (0, 1]$$

H_Q quantifies the combined gravitational and kinematic suppression of internal quantum phase evolution relative to a stationary clock in flat Minkowski space ($H_Q = 1$).

3.3 Surviving Coherent Cycles ($\mathcal{N}_{eff}$) and Stretched-Horizon Regulator (Ξ_{QG})

- **Surviving Coherent Cycles ($\mathcal{N}_{eff}$):** Multiplying proper oscillation frequency f_{local} by relational proper time $t_{pre} = t_{obs} / \rho_\tau$ and Timeline Integrity Φ_τ yields $\mathcal{N}_{eff} = (f_{local} \, t_{pre}) \, \Phi_\tau$, counting physical phase cycles accumulated along the observer's Page–Wootters clock.
- **Stretched-Horizon Membrane Regulator (Ξ_{QG}):** Near the horizon ($r \to r_+$), where classical $N^2(\theta) \to 0$ would freeze relational clock evolution ($\rho_\tau \to \infty$), the stretched-horizon membrane

ratio $\Pi_{horizon} = 10^{-6} / N^2(\theta)$ within Ξ_{QG} enforces the boundary $N^2(\theta) \geq 10^{-6}$ (Thorne, Price, & MacDonald, 1986), keeping relational proper time finite and causal.

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